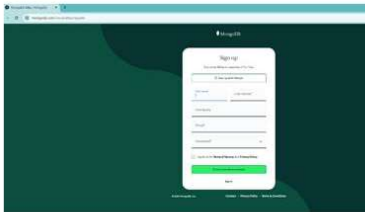


Lab: Setting Up a MongoDB Atlas Account and Free Cluster

MongoDB Atlas offers you free registration to set up a free cluster. In this exercise, you will create an account by executing the following steps:

- Go to <https://www.mongodb.com> and click **Start free**
- You can sign up using your Google account or by providing your details manually as can be seen from the following screen.



- Next, you should see the following page, which means your account has been successfully created:

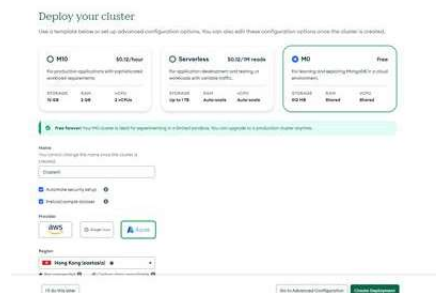


- Enter your organisation and project details.

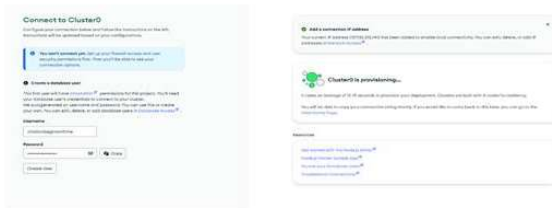


Setting Up Your First Free MongoDB Cluster on Atlas

- Click **Build a Cluster** to configure your cluster. A cluster configuration screen will be presented to select different options for your cluster.
- Select the **Free** cluster for learning. Select the cloud provider of your choice. You can use the default cloud provider. For this exercise, we will be using **Azure**.

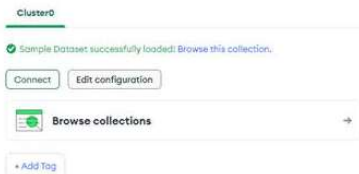


- Ensure that the selections are made correctly in the preceding step so that the cost appears as **FREE**. Any selections different from what is recommended in the previous steps may add costs for your cluster. Click on **Create Cluster**.



- After a few minutes, you should see your new cluster, as shown here

Clusters



- Select "Browse collections." You will be able to see the movie collections which you will use for your lab practices later.



Connecting to your MongoDB Atlas Cluster from Compass by Configuring Network Access and Database User

- Here are the steps to connect to your MongoDB Atlas cluster running on the cloud.
- Click on "Network Access" on the left navigation bar.



- Check if your current IP Address has been whitelisted. Otherwise, click on "Add IP Address" on the top right.



- Select "Add Current IP Address." The screen will show your current IP address after clicking on the Add IP Address button. If you wish to add more IPs to the whitelist, you can add them manually by clicking the Add a Different IP Address option.



- To avoid IP related complications, for now we'll select the third option that says Allow Access from Anywhere. This means Atlas will allow you to access the cluster from any IP address (0.0.0.0)



- Run "cmd" in command prompt. Run "ipconfig" to view your network. Add the IPv4 Address. If you are behind a firewall, e.g. organisation's firewall, you may not be able to access your atlas cluster as certain ports are blocked.

```

Wireless LAN adapter Wi-Fi 2:

Connection-specific DNS Suffix  . : 
Link-local IPv6 Address . . . . . : fe80::7cfa:c0e9:e39d:2a8e%17
IPv4 Address. . . . . : 192.168.1.4
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.254

Ethernet adapter Bluetooth Network Connection 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : 
C:\Users\sowj>ipconfig

```

Network Access

IP Access List Peering Private Endpoint

You will only be able to connect to your cluster from the following list of IP Addresses:

IP Address
192.168.1.4/32
16.14.189.24/32 (includes your current IP address)
0.0.0.0/0 (includes your current IP address)

- Click on Database Access. Select add a new database user. Select password, add a user and password for connection later. Set to built-in role to read and write all databases. Then click add user.

Add New Database User

Create a database user to grant an application or user access to databases and collections in your clusters in this Atlas project. Database access control can be configured with default privileges or custom roles. You can grant access to an Atlas project or organization using the corresponding access manager.

Authentication method

Password **Certificate** **Add user** (MongoDB 4.4 and up) **Anonymous user** (MongoDB 3.0 and up)

MongoDB uses SCRAM-SHA-256 as its default authentication method.

Password authentication

Username: admin

Password: admin

☐ Authenticate Schema Password

Database user privileges

Configure the database access control by designating database user to one of the built-in role, creating custom role, and modifying specific privileges. A user can grant access to all entities within the user designed to them, not just the system database where it is created. You must choose at least one role or privilege. [Learn more about roles](#)

Built-in role

Select one built-in role for the user.

Read and write to all databases

Custom roles

Select your pre-defined custom roles. Create a custom role in the [Custom roles](#) page.

Specific privileges

Select privilege privileges and what database and collection they are associated with. Leaving collection empty will grant this role for all collections in the database.

Restrict Access to Specific Clusters/Filtered Database instances/Stream Processing instances

Enable to restrict the resources that user can access. By default, all resources in this project are accessible.

Temporary user

This user is temporary and will be deleted after your specified duration of 1 hour, 1 day, or 1 week.

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Database Access

Database Users Custom Roles

User Name	Authentication Method	MongoDB Role
admin	SCRAM	readWriteAnyDatabase-Admin

Connecting to your MongoDB Atlas Cluster from Mongo Compass

- Click on "Overview" on the left navigation bar.
- Click on "Connect" under Cluster 0.

Clusters

Cluster0

Connect

Edit configuration



Browse collections

- Click "View full instructions"

Done

View full instructions

- Select the admin database user. Choose developer tools , and Compass (GUI) and select your operating system.

Connect to Cluster0

Configure your connection below and follow the instructions on the left. Instructions will be updated based on your configurations.

Select database user

admin:root

Select client category

Drivers: Access your Atlas data using MongoDB's native drivers (e.g. Node.js, Py, etc.)

Developer Tools: Access your Atlas data through tools like Compass, Shell, VS Code

Choose a tool

MongoDB Shell

Compass (GUI)

MongoDB for VS Code

☐ I have installed Compass

Select your operating system

Windows 64-bit (x64)

- Copy the connection string and open your Mongo Compass.

Customized instructions based on your inputs:

1. Download MongoDB Compass and version

Download Compass (1.43.3) or Copy download URL

Compass is an interactive tool for querying, optimizing, and analyzing your MongoDB data.

2. Copy the connection string, then open MongoDB Compass

mongodb+srv://admin:<password>@cluster0.z0eaeuf.mongod.net/

Replace <password> with the password for the admin user. Ensure any option params are URL encoded.

- Copy the connection string and open your Mongo Compass.

Example: "mongodb+srv://admin<password>@cluster0.z0eaeuf.mongod.net/" Replace <password> with your password

New Connection

Connect to a MongoDB deployment

FAVORITE

URI

mongodb+srv://adminadmin@cluster0.z0eaeuf.mongod.net/

Advanced Connection Options

Save

Save & Connect

Connect

- Upon successful connection, you should be able to see the sample_mflix database from your Compass. You have successfully connect to Atlas (cloud NOSQL database) using the Compass and access the database. Congratulations!

Create collection Refresh

View Sort by Collection Name

comments

Storage size: 5.95 MB	Documents: 41 K	Avg. document size: 284.00 B	Indexes: 1	Total index size: 137 MB
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embedded_movies

Storage size: 71.11 MB	Documents: 3.5 K	Avg. document size: 2164 kB	Indexes: 1	Total index size: 147.46 kB
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movies

Storage size: 19.60 MB	Documents: 21 K	Avg. document size: 1.60 kB	Indexes: 2	Total index size: 17.84 MB
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sessions

Storage size: 20.46 kB	Documents: 1	Avg. document size: 640.00 B	Indexes: 2	Total index size: 40.96 kB
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theaters

My Queries

Performance

Databases

Search

- admin
- local
- sample_mflix
 - comments
 - embedded_movies
 - movies
 - sessions
 - theaters
 - users